



COLLEGE OF  
PAEDIATRICS  
ACADEMY OF MEDICINE OF MALAYSIA

# Guidelines on the Management of **Acute Diarrhoea in Children**

**2026**

**2nd Edition**

# Guidelines on the Management of Acute Diarrhoea in Children 2026

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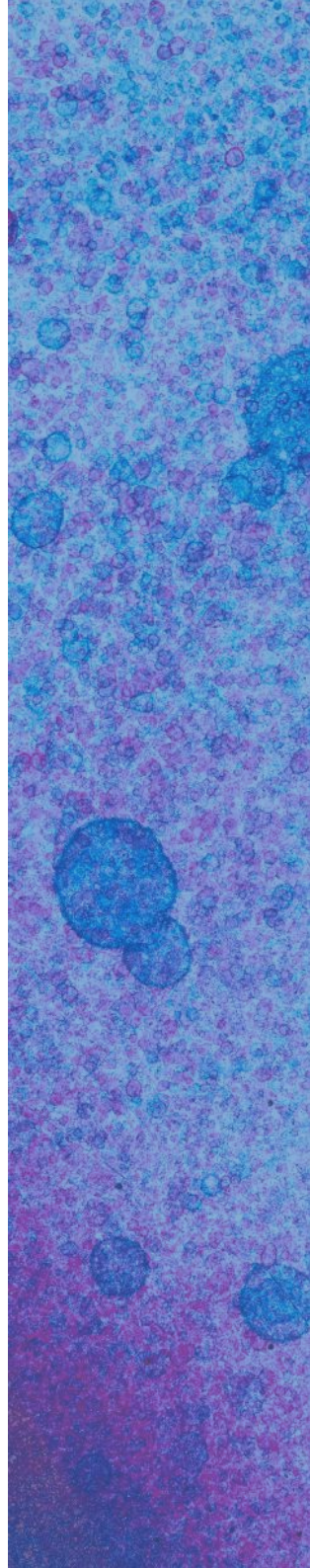
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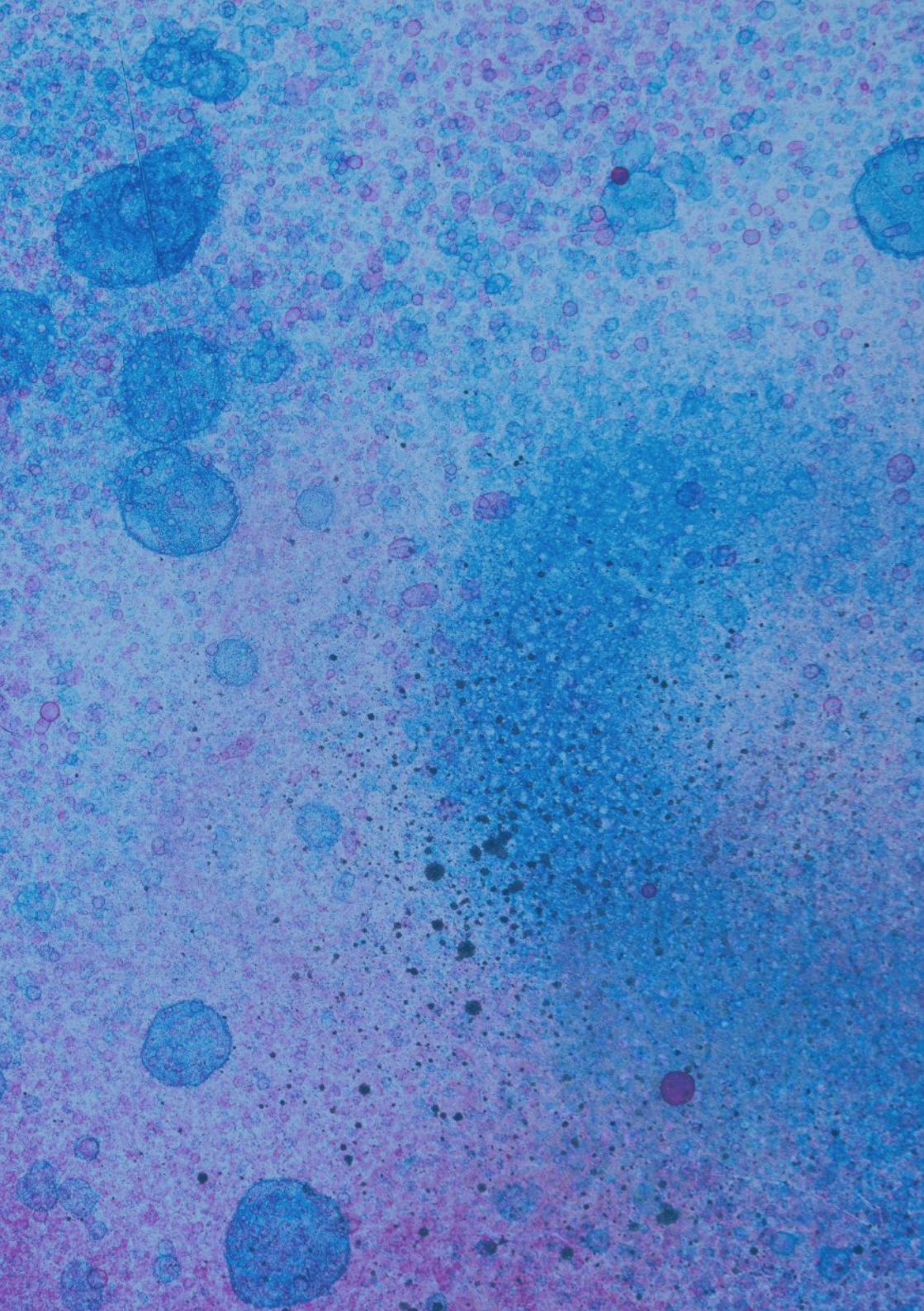
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## **DESIGNED BY:**

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# Committee Members

## Chair

### **Professor Dr Lee Way Seah**

MSEF Chair Professor in Medicine and Senior Professor in Paediatrics  
M. Kandiah Faculty of Medicine and Health Sciences  
Universiti Tunku Abdul Rahman,  
Kajang, Selangor

## Members

### **Dr Alvin Khoh Kim Mun**

Consultant Paediatrician and Paediatric  
Gastroenterologist and Hepatologist  
Gleneagles Hospital Kota Kinabalu,  
Sabah

### **Dr Chew Kee Seang**

Consultant Paediatric  
Gastroenterologist and Hepatologist  
Universiti Malaya Medical Centre,  
Kuala Lumpur

Medical Lecturer  
Department of Paediatrics,  
Faculty of Medicine,  
Universiti Malaya,  
Kuala Lumpur

### **Dr Chong Sze Yee**

Consultant Paediatric  
Gastroenterologist and Hepatologist  
Department of Paediatrics,  
Hospital Raja Permaisuri Bainun,  
Ipoh

### **Dr Choong Chee Liang**

Paediatric Gastroenterologist  
and Hepatologist  
Department of Paediatrics,  
Hospital Pulau Pinang,  
Pulau Pinang

### **Associate Professor**

#### **Dr Erwin Jiayuan Khoo**

Head, Department of Paediatrics,  
School of Medicine,  
International Medical University,  
Kuala Lumpur

Affiliate, Centre for Bioethics,  
Harvard Medical School,  
Harvard University,  
Boston, United States

### **Associate Professor**

#### **Dr Intan Juliana Abd Hamid**

Department of Clinical Medicine,  
Institut Perubatan dan  
Pergigian Termaju,  
Universiti Sains Malaysia,  
Pulau Pinang

**Dr Tee Chun Wei**

Paediatrician  
Universiti Malaya Medical Centre,  
Kuala Lumpur

**Datuk Dr Zulkifli Ismail**

Consultant Paediatrician and  
Paediatric Cardiologist  
KPJ Selangor Specialist Hospital,  
Shah Alam

**TARGET READER:**

The target reader of this guideline includes paediatricians, primary care physicians, and other healthcare professionals involved in the care of children.

**FORMAT:**

This guideline consists of easy-to-read information with appropriate tables. As far as possible, all information is evidence-based. Where evidence is not currently available, a consensus from the combined opinions of the committee members was used.

**AVAILABILITY**

The electronic version of this guideline may be obtained from the following websites: Malaysian Paediatric Association (MPA): <https://mpaeds.my/>

Malaysian Society of Paediatric Gastroenterology, Hepatology and Nutrition (MySPGHAN): <https://www.myspgghan.my/>

College of Paediatrics, Academy of Medicine of Malaysia (AMMCOP): <http://www.acadmed.org.my/>

**CONTENT**

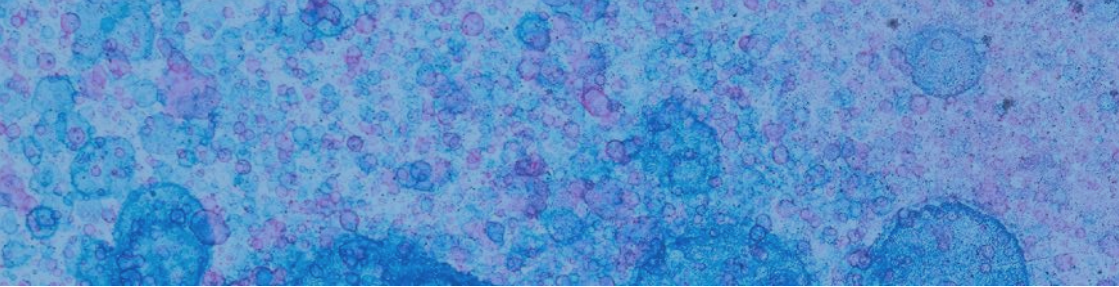
The committee members retain exclusive authority over the guideline's content, ensuring its development remains fully independent and free from any external influence.

**DISCLOSURE**

No conflict of interest is declared by any of the committee members

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|                                                        |           |
|--------------------------------------------------------|-----------|
| <b>1 INTRODUCTION</b>                                  | <b>8</b>  |
| <b>2 SUMMARY</b>                                       | <b>9</b>  |
| <b>3 SCOPE AND SETTING</b>                             | <b>11</b> |
| <b>4 CLINICAL SIGNIFICANCE OF DIARRHOEA</b>            | <b>12</b> |
| <b>5 DEFINITION</b>                                    | <b>13</b> |
| <b>6 CLINICAL TYPES OF DIARRHOEA</b>                   | <b>14</b> |
| <b>7 IMPORTANT CAUSATIVE AGENTS OF GASTROENTERITIS</b> | <b>15</b> |
| <b>8 ASSESSMENT OF ACUTE DIARRHOEA</b>                 | <b>18</b> |
| 8.1 History                                            |           |
| 8.2 Physical Examination                               |           |
| 8.3 Dehydration                                        |           |
| 8.4 Differential Diagnoses                             |           |
| 8.5 Bacterial or Viral Cause?                          |           |
| 8.6 Diagnostic Workup                                  |           |



# TABLE OF CONTENT

|                                                                                   |           |
|-----------------------------------------------------------------------------------|-----------|
| <b>9 MANAGEMENT OF CHILDHOOD ACUTE DIARRHOEA</b>                                  | <b>26</b> |
| 9.1 Referral for Hospital Care                                                    |           |
| 9.2 Rehydration in Acute Diarrhoea                                                |           |
| 9.3 Special Consideration                                                         |           |
| 9.4 Adjunctive Therapy for Acute Diarrhoea                                        |           |
| 9.5 Nutritional Therapy / Special Infant Formula                                  |           |
| 9.6 Return to Childcare or Water Recreation Following Acute Viral Gastroenteritis |           |
| <b>10 PREVENTION OF CHILDHOOD AGE</b>                                             | <b>45</b> |
| 10.1 Vaccines                                                                     |           |
| <b>11 SPECIAL CONSIDERATIONS</b>                                                  | <b>48</b> |
| 11.1 Persistent Diarrhoea (PD)                                                    |           |
| 11.2 Lactose Intolerance                                                          |           |
| 11.3 Diarrhoea in a Malnourished Child                                            |           |
| 11.4 Refeeding Syndrome                                                           |           |
| 11.5 Cow's Milk Protein Allergy                                                   |           |
| 11.6 Diarrhoea in Immunocompromised Children                                      |           |
| 11.7 Antibiotic-Associated Diarrhoea                                              |           |
| 11.8 Economic Impact of Childhood Acute Diarrheal Diseases in Malaysia            |           |
| <b>REFERENCES</b>                                                                 | <b>58</b> |

## 1. Introduction

In 2017, a systematic review on Global Burden of Disease showed that the under-five mortality rate of communicable diseases in Malaysia had reduced fourfold between 1990 and 2013.<sup>1</sup> Mortality caused by non-communicable diseases is now higher than that of communicable diseases.<sup>1</sup> The 2016 National Health and Morbidity Survey, involving 15,188 respondents, showed the prevalence of diarrhoea in the two weeks preceding the survey among children under five years old was 4.4% (95% CI: 3.8, 5.2).<sup>2</sup> The use of untreated water is associated with twice the risk of getting acute gastroenteritis (AGE).<sup>2</sup>

The first edition of the Malaysian Guidelines on the Management of Acute Diarrhoea in Children, published in 2011 and endorsed by the Academy of Medicine, College of Paediatrics (AMMCOP) and Malaysian Paediatric Association (MPA), addressed the unique needs of Malaysian children afflicted with acute diarrhoea. Since then, several advances have been made in the diagnosis, management and prevention of this age-old problem affecting children the world over. This Second Edition updates the previous version and presents the latest evidence-based guidance on the epidemiology, investigation, management, and prevention of acute childhood diarrhoea for all healthcare providers in Malaysia.



## 2. Summary

Acute diarrhoea remains the leading cause of death among children worldwide. In 2021, the World Health Organization (WHO) reported that diarrhoeal disease was responsible for approximately 400,000 deaths among children under 5 years annually, with an additional 50,000 deaths occurring in children aged 5 to 9 years.

In Malaysia, data from the 2016 Under-5 Mortality Review (Ministry of Health, Government of Malaysia; 2022) indicated that out of 5005 deaths among children under 5 years, 53 were attributed to acute gastroenteritis.

Timely and accurate clinical assessment is essential for diagnosing the condition. Every child with acute diarrhoea should be carefully assessed for dehydration and other complications. Criteria for hospital care include moderate-to-severe dehydration, persistent vomiting or worsening of diarrhoea even in the absence of dehydration, uncertainty about diagnosis, presence of unfavourable socio-economic factors, or other complications.

In most cases of uncomplicated acute diarrhoea without significant dehydration, laboratory investigations are unnecessary. A thorough history and physical examination to assess hydration status are usually sufficient.

Oral rehydration therapy is the treatment of choice in most cases of uncomplicated acute diarrhoea. Drug therapy is generally unnecessary and may even be contraindicated or potentially harmful. Several adjunct therapies have been shown to reduce the duration of diarrhoea and may be considered for use in children with acute diarrhoea.

Children who require rehydration should continue breastfeeding or be formula-fed. Food should not be withheld for more than 4 to 6 hours after the onset of rehydration.

Breastfeeding is the most effective preventive measure against acute diarrhoea in young infants. Rotavirus is the most common viral pathogen causing acute dehydrating diarrhoea globally. Although rotavirus vaccines are not part of the national immunisation schedule in Malaysia, vaccination should be considered to help prevent rotavirus-related diarrhoea.

### 3. Scope and setting

#### **Setting**

The present guidelines are intended for use in the following settings:

- Primary care (outpatient clinic, emergency room)
- In-patient care

#### **Focus**

The guidelines focus on the management of acute diarrhoea in children with special attention to Malaysia. The following aspects will be covered:

- Clinical assessment, including assessment of dehydration
- Principles of rehydration therapy
- Adjunctive therapy, including the role of pharmacological agents (e.g., antibiotics, anti-diarrhoeal and antiemetics), probiotics, and prebiotics
- Nutritional management
- Prevention: rotavirus vaccines



### 4. Clinical significance of diarrhoea

Acute diarrhoea causes rapid loss of fluid and electrolytes in infants and young children, leading to dehydration, electrolyte imbalances, and in severe cases, shock and even death. Severe dehydrating diarrhoea is associated with significant morbidity and mortality, especially in infants, young children, and children with underlying co-morbidities and neuro-disability. These vulnerable groups often cannot express the need for additional fluids and have limited physiological capacity to adapt to acute changes.

If the diarrhoea persists, undernutrition and other nutritional deficiencies may ensue, further complicating the subsequent course of the illness. Early recognition of dehydration and instituting appropriate management in a timely manner prevent morbidity and mortality.

Diarrhoea is usually a symptom of an infection of the gastrointestinal tract. However, acute diarrhoea may also be part of clinical features of non-gastrointestinal illnesses such as acute otitis media, pneumonia, meningitis, and acute abdominal emergencies, e.g., appendicitis and intussusception.

## 5. Definition

Diarrhoea is generally defined as the passage of loose or watery stools at least three times in a 24-hour period.<sup>3</sup> Young infants, especially breastfed infants, often pass paste-like stools and should not be considered to have diarrhoea. Exclusively breastfed infants under 4 months old have significantly higher stool frequency compared with formula-fed infants (median [IQR] 2.3 [3.2] vs 1.0 [0.8],  $p < 0.002$ ).<sup>4</sup> Hence, in young infants, diarrhoea should be defined as a change from baseline to more watery stools, with or without increased frequency.

## 6. Clinical types of diarrhoea

| Clinical type                               | Characteristics                                                                                                                                                                                                                                                                              |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acute watery diarrhoea</b>               | It lasts from several hours to a few days, usually less than seven days. The main concerns are dehydration and electrolyte imbalances. Weight loss often reflects fluid loss.                                                                                                                |
| <b>Acute bloody diarrhoea (dysentery)</b>   | It should be considered when blood and mucus are present in the stools. The main dangers are damage to the intestinal mucosa, sepsis, and malnutrition. Dehydration, although not as common as in acute watery diarrhoea, may also occur.                                                    |
| <b>Persistent diarrhoea</b>                 | It is defined as diarrhoea lasting 14 days or longer. It may lead to undernutrition and systemic infection. Dehydration and electrolyte imbalance may or may not be present, depending on the child's adaptive ability and the speed at which diarrhoea develops.                            |
| <b>Diarrhoea with severe undernutrition</b> | This is a serious condition and warrants special attention to assess for severe systemic infection, dehydration, severe electrolyte imbalance, heart failure (cardiomyopathy), and vitamin and mineral deficiencies. Refeeding syndrome is a particular concern with this group of children. |

## 7. Important causative agents of gastroenteritis

In low-to-middle-income countries (LMICs), rotavirus is the leading cause of diarrhoea requiring hospitalisation (attributable fraction [AF] 33.3%) in children under five years. This is followed by *Shigella* species (9.7%), norovirus (6.5%), and adenovirus (5.5%).<sup>5</sup>

In developed countries such as the United States and the United Kingdom, however, norovirus has emerged as the leading cause of acute diarrhoea, especially after the introduction of the rotavirus vaccination.<sup>6</sup> In LMICs that have introduced rotavirus vaccine into their national immunisation programme, such as those in Central and South America, the proportion of hospitalisations attributable to rotavirus was approximately 50% lower with an AF of 20.8% compared to regions which have not implemented universal rotavirus vaccination (AF 42.1%).

In Malaysia, there is a lack of multicentre, comprehensive studies on the aetiological agents causing acute childhood diarrhoea. Most of the published studies are single-centre and hospital-based (Table 1).

These studies showed that the major enteric viruses causing childhood AGE in Malaysia are rotavirus and norovirus. However, a recent local study using a different technique, a multiplex PCR (polymerase chain reaction) technique, which detects the presence of specific nucleic acid sequences (DNA or RNA) in the pathogens, showed that sapovirus and astrovirus are important pathogens as well.<sup>7</sup>

For bacterial gastroenteritis, the most important causative agent is non-typhoidal *Salmonella*, followed by *Escherichia coli*, *Campylobacter* spp., and *Shigella* spp.<sup>8</sup> These bacteria remain important enteric pathogens in recent studies.<sup>7</sup>



**Table 1: Important causative agents of childhood acute gastroenteritis in Malaysia**

| Author                                 | Settings   | Number tested | Percentage of aetiological agent (%) |            |             |             |                  |                            |                            |                      |
|----------------------------------------|------------|---------------|--------------------------------------|------------|-------------|-------------|------------------|----------------------------|----------------------------|----------------------|
|                                        |            |               | Rotavirus                            | Noro-virus | Adeno-virus | Salmo-nella | Shigella         | <i>E.coli</i> <sup>a</sup> | <i>Campy-lobacter spp.</i> | No organism isolated |
| Iyng-karan et al 1980 <sup>9</sup>     | Hospital   | 300           | 19                                   | -          | 4           | 11          | 6                | 10                         | 0                          | 50                   |
| Koe et al 1991 <sup>10</sup>           | Communi-ty | 97            | 9                                    | -          | NT          | 26          | 2                | 1                          | 3                          | 59                   |
| Lee et al 1997 <sup>11</sup>           | Hospital   | 228           | 29                                   | -          | 4           | 10          | 0.4 <sup>b</sup> | 0.4                        | 0                          | 56                   |
| Poo and Lee 2007 <sup>12</sup>         | Hospital   | 288           | 30                                   | -          | NT          | 9           | 0                | 1                          | 0                          | 60                   |
| Tan 2007 <sup>13</sup>                 | Hospital   | 261           | 54                                   | -          | NT          | 6           | 0.4              | 0.4                        | 0                          | 39                   |
| Ka-har-Ba-dor & Lee 2011 <sup>14</sup> | Hospital   | 568           | 28                                   | 9          | 2           | -           | -                | -                          | -                          | 49                   |
| Kuan et al 2015 <sup>15</sup>          | Hospital   | 98            | 7.1                                  | 7.1        | 1           | 17.3        | -                | 11.2                       | 7.1                        | -                    |
| Nor FM et al 2023 <sup>7,d</sup>       | Hospital   | 745           | 8                                    | 10.5       | 2.7         | 16.6        | 1.3              | 24.8 <sup>c</sup>          | 11.9                       | 16.5                 |

<sup>a</sup> Includes enteropathogenic and enterotoxigenic *E. coli*

<sup>b</sup> Mixed infection

NT Not tested

<sup>c</sup> Includes enteroaggregative *E. coli* (EAEC); enteropathogenic *E. coli* (EPEC); Enterotoxigenic *E. coli* (ETEC); Shiga-like toxin-producing *E. coli* (STEC) stx1/ stx2; and *E. coli* O157

<sup>d</sup> This study included an analysis of samples received from a private hospital group (7 centres) in Klang Valley. The test was performed using FilmArray Gastrointestinal Assay. It tested 22 pathogens—13 bacteria, 5 viruses, and 4 parasites. Of 745 requests, 288 (38.7%) were caused by a single enteropathogen (mono-infection), while the remaining 334 (44.8%) were due to more than one agent (co-infection).

Among children requiring hospitalisation, up to 50% of cases are caused by rotavirus. These children may present with severe vomiting and diarrhoea, often leading to severe dehydration.<sup>13,16,17</sup> However, it seems that the percentage of acute diarrhoea caused by rotavirus has declined over the years. This may be due to the increased uptake of the rotavirus vaccine in some parts of Malaysia, although further studies are needed to confirm this observation, as most of the studies were conducted in the Klang Valley.

In an outpatient setting, the main causative agent is non-typhoidal *Salmonella*. Rotavirus accounts for only 10% of the cases.<sup>10</sup> However, this figure is based on a single Malaysian study conducted at a private outpatient clinic in the Klang Valley, with a relatively small patient sample size.

## 8. Assessment of acute diarrhoea

### Aims of assessment

- Identify the presence, degree, and type of dehydration
- Identify the aetiological agent where clinically indicated and feasible
- Identify co-morbidities and complications
- Assess nutritional status
- Ascertain the most appropriate mode of treatment

### 8.1 History

The following aspects should be covered:

- Onset, frequency, quantity, and character of both vomiting (presence of bile, blood) and diarrhoea (presence of blood or mucus)
- Recent oral intake, including breast milk, other fluids, and food
- Urine output: accurate assessment of urine output is often challenging in children, especially in infants. It is particularly difficult in young female infants since watery stools may be mistaken for urine
- Body weight prior to illness is useful (if recorded in earlier anthropometric assessments)
- Associated symptoms (e.g., fever, generally unwell, changes in mental status)
- Past medical history (e.g., chronic illnesses, recent infections, medication used, immunocompromised states)
- Social history (including recent travel, contact with others, and living environment)

## 8.2 Physical examination

The following aspects should be covered:

- Accurate body weight (may not reflect premorbid weight)
- Vital signs (e.g., temperature, heart rate, respiratory rate, blood pressure)
- General condition (e.g., alertness, activity level, hydration status, nutritional status, level of distress)
- Anterior fontanelle in young infants, depressed in significant dehydration
- Eyes (e.g., sunken appearance, presence or absence of tears)
- Mucous membranes—moist or dry
- Respiratory pattern (e.g., rate, effort, and signs of respiratory distress)
- Bowel sounds (e.g., normal, absent)
- Extremities (e.g., perfusion, capillary filling time, pulse volume)
- Skin turgor (anterior abdominal wall for dehydration assessment)
- Inspection of stool (presence of blood or mucus)

\*Sunken eyes: This sign may be better identified with the help from the caregivers

### 8.3 Dehydration

The classification of severity of dehydration is essential for appropriate fluid management. The best measure of dehydration is the percentage loss of body weight.<sup>3</sup> However, the actual premorbid body weight is often unavailable. It should also be emphasised that clinical signs of dehydration are often imprecise.<sup>18, 19</sup> Therefore, repeated assessments are often necessary.

**Table 2: Clinical assessment of hydration status**

| Assess                                   |                                                                                       |                                                                                                                 |                                                                    |
|------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Look</b> at child's general condition | Well, alert                                                                           | Restless or irritable                                                                                           | Lethargic or unconscious                                           |
| <b>Look</b> for sunken eyes              | No sunken eyes                                                                        | Sunken eyes                                                                                                     | Sunken eyes                                                        |
| <b>Pinch</b> skin of abdomen             | Immediate return of skin to baseline after being pinched                              | Sluggish return of skin after being pinched                                                                     | Skin returns to baseline very slowly (>2 secs) after being pinched |
| <b>Classify</b>                          | <b>Mild Dehydration</b><br><5%<br>Dehydrated*<br><i>IMCI: No signs of Dehydration</i> | ≥2 above signs:<br><b>Moderate Dehydration</b><br>5–10%<br>Dehydrated<br><i>IMCI: Some signs of Dehydration</i> | ≥2 above signs:<br><b>Severe Dehydration</b><br>>10%<br>Dehydrated |

*IMCI: Integrated Management of Childhood Illness*

The most useful signs of significant dehydration are<sup>18, 20</sup>:

- Prolonged capillary refill time (normal < 2 seconds)
- Reduced skin turgor
- Abnormal respiratory pattern

**Table 3: Simplified ways of classifying the degree of dehydration**

| Classification                   | Fluid deficit as % of body weight | Fluid deficit in ml/kg of body weight |
|----------------------------------|-----------------------------------|---------------------------------------|
| <b>No signs of dehydration</b>   | <5%                               | <50 ml/kg                             |
| <b>Some signs of dehydration</b> | 5–10%                             | 50–100 ml/kg                          |
| <b>Severe dehydration</b>        | >10%                              | >100 ml/kg                            |

*Adapted from: WHO 2005<sup>3</sup>*

## 8.4 Hypernatraemic dehydration

Hypernatraemia is defined as a serum sodium concentration of  $\geq 150$  mmol/L. Hypernatraemic dehydration in childhood acute diarrhoea often occurs when young infants who are predominantly breastfed receive inadequate breast milk or are given improperly prepared infant formula.

Signs and symptoms of hypernatraemia largely reflect central nervous system dysfunction and are more prominent when the increase in the serum sodium concentration is marked or occurs rapidly (i.e., within a few hours).



Common signs and symptoms in infants include:<sup>21</sup>

- hyperpnoea
- muscle weakness
- restlessness
- a characteristic high-pitched cry
- insomnia
- lethargy and even coma
- convulsions, typically absent except in cases of inadvertent sodium loading or aggressive rehydration
- 'Doughy' skin on examination, although this sign can be difficult to ascertain

Hypernatraemic dehydration as a complication of AGE is now uncommon. In a one-year prospective study on 393 children admitted to a teaching hospital with AGE, the incidence of hypernatraemia was 1.2%, while that of hyponatraemia was 1.0%.<sup>12</sup>

### 8.5 Differential diagnoses<sup>22</sup>

Symptoms of AGE, such as vomiting, diarrhoea, abdominal pain and fever, are often non-specific. Other extra-intestinal conditions can also present with the above symptoms. These conditions include:

- Other infections (e.g., urinary tract infection, otitis media, pneumonia, meningitis, or other forms of sepsis)
- Surgical causes (e.g., acute appendicitis, strangulated hernia, intussusception, and other causes of bowel obstruction)
- Inflammatory bowel disease
- Drugs
- Food allergy
- Diabetic ketoacidosis or inborn error of metabolism
- Any cause of raised intracranial pressure

Always consider other diagnoses if any of the following warning signs are present:<sup>14</sup>

- Abdominal distension
- Bile-stained vomiting
- Blood in vomitus or stool (in an appropriate clinical setting)
- Severe abdominal pain
- Vomiting without diarrhoea
- Headache

### 8.6 Bacterial or viral cause?

It should be emphasised that the clinical features of both viral and bacterial aetiology overlap considerably.<sup>23</sup> Generally, however, the presence of high fever ( $>39^{\circ}\text{C}$ ), overt faecal blood, abdominal pain, and central nervous system symptoms such as irritability, apathy, seizures or coma suggests bacterial aetiology.<sup>23, 24</sup> On the other hand, children presenting with significant vomiting and respiratory symptoms are likely to have a viral aetiology.<sup>25</sup> Rotavirus infection is associated with more severe vomiting and dehydration.

### 8.7 Diagnostic workup<sup>3, 18</sup>

In most children with uncomplicated AGE and no signs of dehydration or only mild-to-moderate dehydration, laboratory investigations are not necessary.<sup>18</sup>

Laboratory tests to assess the severity of dehydration are imprecise.

Electrolytes should be measured in:

- moderately dehydrated children whose history or physical examination findings are inconsistent with a simple diarrhoeal illness and all severely dehydrated children.<sup>18</sup>
- all children requiring intravenous therapy, and during therapy to monitor the presence of hyper- and hyponatraemia and other electrolyte imbalances.<sup>18</sup>

**8.7.1 Stool**

- Stool culture is not routinely indicated as it is expensive and does not alter the management in most children.<sup>18</sup>
- Stool culture should be considered if there is profuse watery diarrhoea (indicative of cholera), or blood and mucus in the stool (suggestive of bacterial dysentery).
- Viral testing is usually not indicated.

**8.7.2 Stool PCR<sup>26, 27</sup>**

The stool PCR (polymerase chain reaction) test is a molecular diagnostic technique used to detect the presence of specific nucleic acid sequences of pathogens (e.g., bacteria, viruses, or parasites) in stool samples. The test relies on the amplification of target DNA or RNA to identify infectious agents with high sensitivity and specificity. While PCR has substantially improved the diagnostic yield compared to conventional culture-based methods, its high sensitivity may lead to the detection of non-viable or colonising organisms, potentially resulting in false-positive interpretations. This poses a particular challenge in establishing the aetiological role of detected pathogens, especially in cases involving asymptomatic carriage or co-infection with multiple organisms. Therefore, the interpretation of stool PCR results requires careful clinical correlation to distinguish true pathogenic involvement from incidental findings.<sup>28</sup>

**8.7.3 Indications for stool culture and/or PCR**

Most mild cases of childhood acute diarrhoea do not require diagnostic testing and can be managed with supportive care such as hydration and monitoring. Stool testing, including culture and/or PCR, is typically reserved for cases with certain indications:

- Severe symptoms: Bloody or mucoid diarrhoea, high fever, severe abdominal pain, or signs of systemic infection.
- Prolonged illness: Diarrhoea lasting more than 7–10 days.

- Immunocompromised children: Those with weakened immune systems may require stool testing to identify specific pathogens.
- Outbreak situations: For children who are part of a group (e.g., daycare) with multiple cases of gastroenteritis or during known outbreaks.
- Travel history: Recent travel to areas where specific pathogens (e.g., Salmonella, Shigella, E. coli, or intestinal parasites) are endemic.
- Hospitalisation: For hospitalised children with severe dehydration or complications.

### **8.7.4 Urine**

- Urine microscopy should not be performed routinely due to the risk of sample contamination during episodes of acute diarrhoea.

### **8.7.5 Blood**

- Consider testing for urea,  $\text{Na}^+$ ,  $\text{K}^+$ , creatinine, pH,  $\text{HCO}_3^-$
- Perform a complete blood count if bacterial sepsis is suspected.
- Consider glucose monitoring, especially for young children who have vomiting.
- Consider  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  in young infants.

The incidence of hypoglycaemia in children with AGE is estimated to be between 6 and 11%.<sup>29–32</sup> Risk factors for developing hypoglycaemia are rotavirus infection, bacteraemia, frequent vomiting, and undernutrition.<sup>29–32</sup>

## 9. Management of acute childhood diarrhoea

Key considerations include:

- Prevention and correction of dehydration and electrolyte imbalance.
- Identification of children at risk of severe disease or developing complications.
- Prevention and treatment of complications such as invasive disease, severe electrolyte imbalance, metabolic disturbances, and undernutrition.
- Provision of adequate and appropriate nutrition.
- Use of medications, which may have a supplementary role.

### 9.1 Referral for hospital care

For uncomplicated childhood AGE with no significant dehydration, outpatient assessment and subsequent home management are usually sufficient.<sup>3, 18</sup> The following warning signs can help determine whether further assessment and possible hospital care are required<sup>18, 33</sup>:

- Severe dehydration (>10% of body weight), or signs of shock
- Neurological abnormalities (e.g., lethargy, seizures, etc)
- Persistent or bilious vomiting, even in the absence of dehydration
- Failure of oral rehydration salts (ORS) therapy
- Presence of systemic illness (e.g., high fever, toxic appearance)
- Underlying medical conditions (e.g., heart failure, chronic kidney disease, short bowel syndrome)
- Inability of caregivers to provide adequate care at home, or other social/logistic concerns
- Suspected surgical condition, or uncertainty about the diagnosis
- Uncertain assessment of dehydration severity (e.g., in severely obese or undernourished children)

However, it should be emphasised that there are currently no established, evidence-based criteria for the hospital admission of childhood AGE.<sup>18, 33</sup>

## 9.2 Rehydration in Acute Diarrhoea

General principles: Oral rehydration should generally be used as first-line therapy for the management of children with AGE.<sup>3, 18, 34, 35</sup> When oral rehydration is not feasible, enteral rehydration via the nasogastric route is equally effective—if not superior—to intravenous rehydration.<sup>3, 18, 34-36</sup> Oral or enteral rehydration, compared with intravenous therapy, is associated with significantly fewer major complications and a shorter hospital stay. It is successful in most children.<sup>33</sup>

### **9.2.1 When there are no signs of dehydration (<5%):**

- If there is no excessive vomiting, home management is usually sufficient.
- Continue breastfeeding if the child is breastfed.
- Continue usual feeding if the child is formula-fed and offer additional fluids (e.g., ORS).
- For older children, continue their normal diet and provide extra fluids (e.g., ORS).

### **9.2.2 When there are some signs of dehydration (5–10%):**

- If no excessive vomiting and in the absence of adverse social circumstances, outpatient therapy can still be considered.
- Administer ORS 50–100ml/kg over 4 hours.
- After each episode of diarrhoea/vomiting, administer ORS 10ml/kg.
- Provide small and frequent feeds with regular assessment.
- Refer for hospital admission and intravenous fluids if vomiting persists or dehydration worsens despite adequate therapy.

**9.2.3 When there are signs of severe dehydration (>10%):**

- Initiate resuscitation with normal saline or Ringer's lactate.
- Monitor the child frequently.
- Refer immediately to hospital for admission.

**9.2.4 Selection of oral rehydration salts (ORS)**

Table 4 shows the electrolyte content of stool samples obtained from patients with cholera, rotavirus, and enteropathogenic *Escherichia coli*. The original WHO ORS (1975 formulation) is more appropriate in situations where cholera is likely. In other situations, a reduced osmolality ORS is preferred.

**Table 4: Mean stool Na<sup>+</sup> and K<sup>+</sup> (mmol/L) concentrations according to duration of diarrhoea**

|          | Cholera  |          | EPEC     |          | Rotavirus |          |
|----------|----------|----------|----------|----------|-----------|----------|
| Duration | Na       | K        | Na       | K        | Na        | K        |
| (h)      | (mmol/L) | (mmol/L) | (mmol/L) | (mmol/L) | (mmol/L)  | (mmol/L) |
| 0–12     | 98       | 29       | 67       | 37       | 53        | 46       |
| 13–24    | 83       | 37       | 55       | 38       | 42        | 42       |
| 25–48    | 63       | 28       | 44       | 26       | 32        | 28       |
| 48+      | 46       | 65       | 44       | 37       | 34        | 43       |

Adapted from Molla AM et al, 1991 (37)

EPEC: enteropathogenic *Escherichia coli*



Table 5 shows the electrolyte content of various ORS preparations available in Malaysia. Fluids that are not suitable as substitutes for ORS include 100 Plus (Table 6).

**Table 5: Electrolyte composition, osmolality of various ORS preparations**

|                               | Na<br>(mmol/L) | K<br>(mmol/L) | Cl<br>(mmol/L) | HCO <sub>3</sub> <sup>-</sup><br>(mmol/L) | Glucose<br>(mmol/L) | Citrate #<br>(mmol/L) | Osmo-<br>lality<br>(mmol/<br>kg) |
|-------------------------------|----------------|---------------|----------------|-------------------------------------------|---------------------|-----------------------|----------------------------------|
| Standard WHO (1975)           | 90             | 20            | 80             | 30                                        | 111                 |                       | 311                              |
| Reduced-Osmolality WHO (2002) | 75             | 20            | 65             | 30                                        | 75                  |                       | 245                              |
| ORS B Plus (per sachet)       | 75             | 20            | 65             |                                           | 75                  | 10                    | 245                              |
| Enerlyte                      | 75             | 20            | 65             |                                           | 75                  | 10                    | 245                              |
| Restors Oral Powder           | 74             | 20            | 65             |                                           | 75                  | 10                    | 244                              |
| N-ORS                         | 75             | 20            | 65             |                                           | 75                  | 10                    | 245                              |
| Oralite                       | 75             | 20            | 65             |                                           | 75                  | 10                    | 245                              |
| Uphalyte                      | 56             | 20            | 56             | 20                                        | 139                 |                       |                                  |

Source: WHO(3); National Pharmaceutical Regulatory Agency (NPRA) website product search, accessed 24/3/2025

# one molecule of citrate is metabolised into 3 molecules of bicarbonate in the body.

**Table 6: Fluids not appropriate as rehydration therapy**

|                                         | Na<br>(mmol/L) | K<br>(mmol/L) | Cl<br>(mmol/L) | HCO <sub>3</sub> <sup>-</sup><br>(mmol/L) | Glucose<br>(mmol/L) | Osmolality<br>(mmol/kg) |
|-----------------------------------------|----------------|---------------|----------------|-------------------------------------------|---------------------|-------------------------|
| Coca cola<br>(Classic)                  | 0.27           | NA            | NA             | NA                                        | 83*                 | 695 (38)                |
| Apple juice                             | 3              | 20            | NA             | 0                                         | 600–900             | 718                     |
| Coconut<br>water<br>(young)<br>(39, 40) | ~33            | ~51           | ~52            | NA                                        | 55.5                | 288                     |
| Chicken<br>broth                        | 250            | 8             | NA             | 0                                         | 0                   | 260                     |
| Tea                                     | 0              | 0             | NA             | 0                                         | 0                   | 31                      |
| 100 Plus                                | 19.5           | 3.1           | 11             | 3                                         | 272                 | 330                     |

*Adapted from Guidelines on the Management of Acute Diarrhoea in Children 2011, Feldman M (38), Chavalittamrong B (39), Kalman DS (40), \*Coca cola Malaysia official website, 100 plus product label*

NA Not available

## 9.3 Special consideration

### 9.3.1. Hypernatraemic dehydration

Hypernatraemic dehydration is defined as a serum sodium concentration of more than 150 mmol/L. Generally, patients with hypernatraemic dehydration respond well to oral rehydration therapy.<sup>34</sup> Those with severe dehydration should first receive intravenous fluid therapy for resuscitation. An isotonic solution (normal saline) should be used for subsequent rehydration, administered slowly over 48 hours. ORS may also be used to continue rehydration. ORS is safer than intravenous fluids because it is less likely to lead to a rapid increase in intracellular water, which may lead to seizures and raised intracranial pressure.<sup>34</sup>

## 9.4 Adjunctive therapy for acute diarrhoea

### 9.4.1 Antibiotics

The routine use of antibiotics for diarrhoea in children is not recommended with the exception of clinically recognizable severe cases. (3) The majority of gastroenteritis cases in children are viral in origin, caused by rotavirus, norovirus, and adenovirus. Therefore, antibiotics should be reserved for specific pathogens or well-defined clinical indications.

#### **Recommendations:**

Antibiotics are indicated in the following situations:

- Shigella dysentery: cases presenting as bloody diarrhoea should be treated with an antimicrobial effective for *Shigella*.
- When cholera is suspected.
- When diarrhoea is associated with another acute infection such as pneumonia or urinary tract infection.
- Antibiotics may be indicated for *Salmonella* gastroenteritis in very young infants (<3 months), or in patients who are immunocompromised or immunosuppressed, systemically ill, or have achlorhydria (e.g., due to proton pump inhibitor therapy).

**Table 7: Recommended antibiotics for specific causes of diarrhoea<sup>3, 33, 41</sup>**

| Pathogen                                                                   | Suggested Treatment                                              |                                                                                         | Notes                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                            | Recommended                                                      | Alternative(s)                                                                          |                                                                                                                                                                                                                                                                                    |
| Acute Watery Diarrhoea                                                     |                                                                  |                                                                                         |                                                                                                                                                                                                                                                                                    |
| Rotavirus, norovirus, other viruses                                        | Antibiotics are not indicated                                    |                                                                                         |                                                                                                                                                                                                                                                                                    |
| Cholera                                                                    | PO azithromycin 20 mg/kg (max 1 g) single dose                   | PO doxycycline 2–4 mg/kg (maximum 1 g) single dose (>8 years)                           |                                                                                                                                                                                                                                                                                    |
| Dysentery                                                                  |                                                                  |                                                                                         |                                                                                                                                                                                                                                                                                    |
| <i>Shigella</i> spp., <i>Campylobacter</i> spp., <i>Escherichia coli</i> * | <b>Mild, uncomplicated:</b>                                      | Consider oral antibiotics if there is the presence of high-grade fever and bloody stool | <i>Campylobacter</i> spp.: Macrolides (azithromycin) is preferred<br><br>*There are reports demonstrated that the use of $\beta$ -lactam antibiotics in children infected with <i>E. Coli</i> is associated with increased risk of haemolytic uraemic syndrome (HUS) <sup>42</sup> |
|                                                                            | Antibiotics are not recommended                                  | PO azithromycin 10 mg/kg for 3–5 days                                                   |                                                                                                                                                                                                                                                                                    |
|                                                                            | <b>Severe, complicated:</b>                                      | PO or IV ampicillin 25 mg/kg/dose q6H for 5–7 days                                      |                                                                                                                                                                                                                                                                                    |
|                                                                            | IV ceftriaxone 75–100 mg/kg/dose q24H (max 2 g daily) for 5 days |                                                                                         |                                                                                                                                                                                                                                                                                    |

| Typhoid fever                                                  |                                                                                                                                      |                                                                                                                                       |                                                                                                        |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <i>Salmonella typhi</i><br><i>S. paratyphi A</i><br>& <i>B</i> | <b>Mild,<br/>uncomplicated:</b><br><br>PO azithromycin<br>10 mg/kg/day for<br>5–7 days                                               | PO ciprofloxacin<br>20 mg/kg/dose<br>(maximum 750<br>mg) q12H for 7<br>days                                                           | Bacteraemia<br>and severe or<br>complicated<br>enteric fever<br>should be<br>treated for 10–14<br>days |
|                                                                | <b>Severe,<br/>complicated:</b><br><br>IV ceftriaxone<br>75–100 mg/kg/<br>dose (max 2g/<br>day) q24H for<br>10–14 days <sup>43</sup> | IV ampicillin 50<br>mg/kg/dose q6H<br><br>OR<br><br>IV ciprofloxacin<br>15 mg/kg/dose<br>q12H for 7–10<br>days                        |                                                                                                        |
| Antibiotic-associated diarrhoea                                |                                                                                                                                      |                                                                                                                                       |                                                                                                        |
| <i>Clostridium<br/>difficile</i>                               | PO<br>metronidazole 30<br>mg/kg/day (max:<br>500 mg/dose) for<br>10 days                                                             | PO vancomycin<br>40 mg/kg/day<br>(max: 125 mg<br>per dose in<br>mild infection,<br>500 mg/dose in<br>severe infection)<br>for 10 days |                                                                                                        |

Note: IV – intravenous, PO – per oral

**9.4.2 Anti-emetics<sup>44, 45</sup>**

The routine use of anti-emetics is not recommended in children with AGE. Children with persistent vomiting should be offered small, frequent feeds.

- Domperidone, metoclopramide, and dimenhydrinate are not effective in stopping vomiting and preventing hospitalization. They may cause sedation and can interfere with oral rehydration therapy. Significant adverse events have also been reported. Metoclopramide is associated with extrapyramidal reactions (e.g., oculogyric crisis) which occur frequently at standard doses. Adverse cardiac events have been reported in children given domperidone.

**Recommendations:**

Domperidone, metoclopramide, and dimenhydrinate are not recommended for the treatment of vomiting in children with AGE.

- Oral ondansetron (as a single dose) has been shown to be safe and effective in children with gastroenteritis, from the age of 6 months. It is effective in reducing vomiting, preventing hospitalisation and reducing the need for intravenous rehydration. The recommended dosage is 0.15 mg/kg. A simplified weight-based dosage regimen may also be used.

| Weight   | Ondansetron dosage |
|----------|--------------------|
| 8–15 kg  | 2 mg               |
| 15–30 kg | 4 mg               |
| >30 kg   | 6–8 mg             |

**Recommendations:**

Ondansetron may be considered for children with persistent vomiting due to acute gastroenteritis.

### 9.4.3 Anti-diarrhoeal agents

- Anti-diarrhoeal agents are generally not indicated in childhood AGE.
- Various anti-diarrhoeal agents and other therapies available in the local market include:

#### 9.4.3.1 Anti-motility (intestinal transit inhibitor; opiate agonists)

##### i. Loperamide

- Loperamide may reduce the frequency and duration of diarrhoea but it is associated with serious side effects, particularly in young children (<3 years). Reported adverse effects include lethargy and even mortality<sup>46</sup>

##### **Recommendations:**

Loperamide is not recommended for use in children with acute gastroenteritis.

##### ii. Diphenoxylate HCl

- Diphenoxylate is an anti-peristaltic, which masks water loss and prolongs intestinal transit time. It increases direct contact between the intestinal epithelium and noxious agents.
- Its use in *Shigella* infection has been associated with invasive disease, prolonged fever and prolonged excretion of *Shigella*.
- It has a narrow therapeutic range; therefore, the risk of overdose and severe side effects is significant.

##### **Recommendations:**

Diphenoxylate is not recommended for use in children with acute gastroenteritis.

### 9.4.3.2 Intraluminal agents (adsorbents, bulk-forming, etc)

#### i. **Silicates: kaolin/pectin**<sup>47</sup>

- Binds microorganisms or their products
- Increases stool consistency but does not reduce water and electrolyte loss
- Overall efficacy is limited
- May interfere with the intestinal absorption of nutrients and drugs due to its adsorptive properties

#### **Recommendations:**

Kaolin/pectin is not recommended for the treatment of acute gastroenteritis in children.

#### ii. **Silicates: diosmectite (Smecta®)**

Diosmectite binds to selected bacterial pathogens and rotavirus and has been shown to restore the integrity of damaged intestinal epithelium. It also reduces both stool output and the duration of diarrhoea and has demonstrated effectiveness in rotavirus-associated diarrhoea. It may be used as an adjunct to ORS therapy.<sup>47, 48</sup>

Smecta® is currently licensed to be used in children 2 years and above. It should not be given to mothers who are breastfeeding.

#### **Recommendations:**

May be considered as an adjunctive treatment in children aged 2 years and above.



### 9.4.3.3 Anti-secretory

#### i. Enkephalinase inhibitors: racecadotril

- Racecadotril is an enkephalinase inhibitor that preserves endogenous enkephalins.
- The evidence for routine use is not strong even with two meta-analyses<sup>49, 50</sup> due to heterogeneous study designs and limited data quality. However, racecadotril is not associated with increased adverse events compared to placebo.<sup>48-50</sup>
- Racecadotril appears to be safe and has shown good efficacy against diarrhoea. It can be used as an adjunct to oral rehydration therapy.<sup>50</sup>

#### **Recommendations:**

Can be considered as an adjunct to ORS, but the evidence for efficacy remains weak. Further studies are necessary.

### 9.4.4 Probiotics and Prebiotics<sup>51-53</sup>

#### **Probiotics**

- Certain strains of probiotics may be useful as an adjunct to the management of diarrhoea. Effects of probiotics are both strain-specific and dose-specific. Not all probiotics are equivalent in efficacy, and the effectiveness of a specific strain cannot be generalised.
- Current evidence supports the use of the following probiotic strains as adjuncts in the treatment of acute diarrhoea in children:<sup>54, 55</sup>
  - » *Lactacaseibacillus rhamnosus* GG (*L. rhamnosus* GG)
  - » *Saccharomyces boulardii*
  - » *Limosilactobacillus reuteri* DSM 17938

- *Lactacaseibacillus rhamnosus* GG (*L. rhamnosus* GG)
  - » Combined data from 19 randomized controlled trials (RCTs) on 4073 children showed that *L. rhamnosus* GG significantly reduced the duration of diarrhoea compared with placebo or no treatment (mean difference, -24.02 hours, 95% CI -36.58, -11.45).<sup>56</sup>
  - » The current recommended dose is  $\geq 10^{10}$  CFU/day, for 5 to 7 days.
- *Saccharomyces boulardii*
  - » A meta-analysis of 23 RCTs involving 3450 children showed that *Saccharomyces boulardii* significantly reduced diarrhoea duration (mean difference -1.06 days, 95% CI -1.32 to -0.79).<sup>55</sup>
  - » The current recommended dose is 250 mg–750 mg/day, for 5 to 7 days.
- *Limosilactobacillus reuteri* DSM 17938
  - » A meta-analysis of 4 RCTs involving 347 children showed that *Limosilactobacillus reuteri* DSM 17938 reduced the duration of diarrhoea compared with placebo (mean difference -0.87 days, 95% CI -1.4 to -0.3).<sup>57</sup>
  - » The current recommended dose is  $1 \times 10^8$  to  $4 \times 10^8$  CFU/day for 5 days.
- Currently there are no recommendations for the use of other probiotic strains or products containing multiple strains of probiotics in the treatment of acute childhood diarrhoea.
- There are also no recommendations for the use of prebiotics in acute childhood diarrhoea.

### Recommendations:

*Lactacaseibacillus rhamnosus* GG, *Saccharomyces boulardii* and *Limosilactobacillus reuteri* DSM 17938 may be considered as adjunctive therapies to oral rehydration therapy in the management of acute diarrhoea in children.

### Prevention of acute gastroenteritis

- Currently there is no evidence to support the use of prebiotics or probiotics to prevent AGE.
- There is, however, evidence that certain strains of probiotics may be used to prevent antibiotic-associated diarrhoea. See section titled “Special considerations”.

### 9.4.5 Summary of recommendations (Adjunctive therapy; Table 8)

**Table 8: Adjunctive therapy in childhood acute diarrhoea**

| Adjunctive Therapy  | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Antibiotics</b>  | <p>Indicated in the following situations:</p> <ul style="list-style-type: none"> <li>• <i>Shigella</i> dysentery: In cases presenting with bloody diarrhoea, treatment with an antimicrobial effective against <i>Shigella</i> is indicated.</li> <li>• When cholera is suspected.</li> <li>• When diarrhoea is associated with another acute infection, such as pneumonia or urinary tract infection. May be indicated for <i>Salmonella</i> gastroenteritis in very young infants (&lt;3 months), or in patients who are immunocompromised, immunosuppressed, systemically ill, or have achlorhydria (e.g., due to proton pump inhibitor therapy).</li> </ul> |
| <b>Anti-emetics</b> | <ul style="list-style-type: none"> <li>• Domperidone, metoclopramide, dimenhydrinate — not recommended.</li> <li>• Ondansetron — may be considered in children with persistent vomiting.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Adjunctive Therapy   | Recommendations                                                                                                                                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anti-motility        | <ul style="list-style-type: none"> <li>• Loperamide — not recommended.</li> <li>• Diphenoxylate HCl — not recommended.</li> </ul>                                                                                                                                                                      |
| Intra-luminal Agents | <ul style="list-style-type: none"> <li>• Kaolin/pectin — not recommended.</li> <li>• Diosmectite (Smecta®) — may be considered as an adjunctive treatment.</li> </ul>                                                                                                                                  |
| Anti-secretory       | <ul style="list-style-type: none"> <li>• Enkephalinase inhibitors (racecadotril) — may be considered, but the evidence for efficacy is weak. Further studies are necessary.</li> </ul>                                                                                                                 |
| Probiotics           | <ul style="list-style-type: none"> <li>• <i>Lactocaseibacillus rhamnosus</i> GG, <i>Saccharomyces boulardii</i> and <i>Limosilactobacillus reuteri</i> DSM 17938 — may be considered as an adjunctive therapy to oral rehydration therapy in the management of acute diarrhoea in children.</li> </ul> |

### 9.5 Nutritional therapy / Special Infant Formula

Children who require rehydration should continue to be fed as usual. Food should not be withheld for more than 4–6 hours after the onset of rehydration. Breastfeeding should be continued during episodes of AGE. For formula-fed children, formula dilution and gradual reintroduction of feeding are not recommended.<sup>58</sup>

#### 9.5.1 Undiluted vs. diluted formula

A meta-analysis (1994) investigated the practice of diluting formula 2- to 6-fold for periods ranging from 1–6 days.<sup>58</sup> Children given undiluted formula had a slight increase in stool frequency ( $p = 0.046$ ) compared with those fed diluted formula, but there was no difference in the duration of diarrhoea.

A systematic review (9 RCTs, 687 children) found that diluting lactose-containing formula (by 25%–50%) reduced treatment failure compared with full-strength formula: 11% versus 17% over 24 to 72 hours (NNT=17).<sup>3</sup> However, there was no difference in the duration of diarrhoea or changes in weight.<sup>33, 59</sup>

Diluted lactose-containing milk has not been shown to reduce the duration of diarrhoea compared with undiluted milk or milk products, but it may reduce the risk of treatment failure by around one-third (RR 0.65, 95% CI 0.45 to 0.94, nine trials, 687 participants, low quality evidence).

#### **Recommendations:**

- In children who are on infant formula, dilution of formula is not recommended.

### **9.5.2 Soy-based or cow milk-based lactose-free formula**

Most young children with AGE can safely continue to have lactose-containing milk formula.<sup>58</sup>

Lactose-free products may reduce the duration of diarrhoea by approximately 18 hours (mean difference: –17.77, 95% CI –25.32 to –10.21, 16 trials, 1467 participants; low quality evidence) compared with lactose-containing milk, milk products, or foods. Lactose-free products may also reduce the risk of treatment failure—typically defined as continued or worsening diarrhoea or vomiting, the need for additional rehydration therapy, or on-going weight loss—by around a half (RR 0.52; 95% CI 0.39 to 0.68).<sup>59</sup>

#### **Recommendations:**

A change to lactose-free formula (whether soy-based or cow's milk-based) after an episode of acute diarrhea is not recommended.

### 9.5.3 Zinc

In areas with moderate-to-high risk of zinc deficiency (e.g., Indian subcontinent, sub-Saharan Africa), the United Nations Children's Fund (UNICEF) and the World Health Organization (WHO) recommend zinc supplementation (10 mg below 6 months of age, 20 mg in older infants and children, for 10–14 days) as a universal treatment for children with diarrhoea.<sup>60</sup> This applies to children over 6 months of age with suspected or confirmed zinc deficiency.

A systematic review<sup>58</sup> in children older than 6 months found:

- zinc supplementation may shorten the duration of diarrhoea by approximately half a day.
- it probably reduces the number of children whose diarrhoea persists until day 7.
- the effect appears to be greater in children with signs of malnutrition.

In children younger than 6 months:

- zinc supplementation may have no effect on the mean duration of diarrhoea.
- it may not reduce the number of children with diarrhoea persisting on day seven.
- none of the included trials reported serious adverse events; however, zinc supplementation increased the risk of vomiting in both age groups.

Trials on zinc conducted in populations with a low risk of zinc deficiency (e.g., Europe, Australia) have generally shown no significant benefit. In Malaysia, a review by the Health Technology Assessment Section of the Ministry of Health (2016) found that in children under 5 years old, zinc supplementation shortened the duration of acute diarrhoea and reduced both stool output and frequency.<sup>61</sup> The effects appeared to be greater in children with signs of moderate malnutrition.

However, the evidence for its efficacy in infants under 6 months remains unclear. Overall, zinc supplementation was found to be safe and more cost-effective than standard treatment.

#### **Recommendations:**

- Zinc supplementation may be considered for malnourished children with acute diarrhoea.
- It may also be beneficial in children suspected to have zinc deficiency (e.g., an inadequate dietary intake of zinc-rich foods or clinical signs such as acral dermatitis).

#### **9.5.4 Summary of recommendations (Nutritional therapy)**

| Nutritional Therapy  | Recommendations                                                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infant Formula       | Dilution of formula is not recommended.                                                                                                                                                         |
| Lactose-free Formula | A change to lactose-free formula (either soy-based or cow's milk-based) after an episode of acute diarrhoea is not recommended.                                                                 |
| Zinc                 | Zinc supplementation may be considered in undernourished children with acute diarrhoea. It may also benefit children living in areas with a high prevalence of malnutrition or zinc deficiency. |

### 9.6 Return to childcare or water recreation following acute viral gastroenteritis

Asymptomatic infections play a significant role in the transmission of viral gastroenteritis. Viral excretion may begin before the onset of symptoms and can persist even after symptoms have resolved. When determining whether a child may return to childcare (including nurseries and similar settings) or school after an episode of AGE, the key criteria are the resolution of diarrhoea and the absence of vomiting.

#### **9.6.1 Guidelines for returning to childcare or school**

Children with diarrhoea may attend childcare or school if:<sup>62</sup>

- For infants or toddlers: Stools are contained in the diaper.
- For older children: The child has no accidents using the toilet.
- It has been at least 48 hours since the last episode of diarrhoea or vomiting.<sup>63</sup>

#### **9.6.2 Guidelines for water-related activities**

Children with diarrhoea should avoid swimming and other water-related activities for at least 48 hours from the last episode of diarrhoea or vomiting.

Children with AGE who are not toilet trained should avoid swimming for one week after diarrhoea has resolved.



## 10. Prevention of Childhood AGE

### 10.1 Vaccines

Various enteric vaccines have been developed to reduce the incidence of acute diarrhoea, including vaccines against cholera, typhoid, hepatitis A, and rotavirus.<sup>64–66</sup>

#### **Cholera vaccine**

The efficacy of the currently available oral killed whole-cell cholera vaccines ranges between 50% and 60%.<sup>64</sup> The inclusion of cholera vaccine in immunisation programmes in endemic countries depends on the prevalence of cholera, frequency of outbreaks, and accessibility of healthcare services.<sup>64</sup> Currently, routine immunisation against cholera is not recommended in Malaysia.

#### **Rotavirus vaccine**

Group A rotavirus (RV-A) is the leading cause of severe diarrhoeal disease in infants and young children worldwide.<sup>67</sup> Since 2006, two rotavirus vaccines — RotaTeq® & Rotarix® — have been introduced in Malaysia and are available as optional vaccines.<sup>68</sup> Both have been shown to be highly efficacious in preventing severe RV-A gastroenteritis caused by common RV-A genotypes.<sup>69</sup>

Infants with pre-existing gastrointestinal conditions (e.g., congenital malabsorption syndromes, Hirschsprung's disease, short-gut syndrome), who are not undergoing immuno suppressive therapy should receive rotavirus vaccine.<sup>70</sup>

**Table 9. Recommended schedule for rotavirus vaccines****Recommended Schedule for Rotavirus Vaccines**

| <b>Dose</b> | <b>Rotateq® Age</b> | <b>Rotarix® Age</b> |
|-------------|---------------------|---------------------|
| Primary 1   | 2 months            | 2 months            |
| Primary 2   | 4 months            | 4 months            |
| Primary 3   | 6 months            | -                   |

**Dosage Intervals and Ages for Rotavirus Vaccines**

|                            | <b>RV5 (Rotateq®, Merck)</b> | <b>RV1<br/>(Rotarix®, GSK)*</b> |
|----------------------------|------------------------------|---------------------------------|
| Number of doses in series  | 3                            | 2                               |
| Recommended ages for doses | 2, 4, and 6 months           | 2 and 4 months                  |
| Minimum age for first dose | 6 weeks                      |                                 |
| Maximum age for first dose | 14 weeks 6 days              |                                 |
| Interval between doses     | 4 weeks or more              |                                 |
| Maximum age for last dose  | 8 months 0 days              |                                 |

Note: The use of brand names is not meant to preclude the use of other comparable licensed vaccines.

\* Updated 11/29/2022 to include a new, oral dosing applicator only presentation of Rotarix licensed on 11/4/2022.

Contraindications to rotavirus vaccination include:

1. Serious allergic reaction to vaccine components:
  - a. A severe allergic reaction (e.g., anaphylaxis) following a previous dose of rotavirus vaccine or to a vaccine component.
  - b. Rotarix® oral applicator contains latex rubber; therefore, infants with a severe (anaphylactic) allergy to latex should not receive Rotarix®. RotaTeq® dosing tube is latex-free.<sup>71</sup>
2. Children who are immunosuppressed:
  - a. Infants with known or suspected inborn errors of immunity (e.g., severe combined immunodeficiency).
  - b. Children undergoing chemotherapy for malignancy.

## 11. Special Considerations

### 11.1 Persistent diarrhoea (PD)

The prevalence of persistent diarrhoea (PD) in childhood is unknown and varies significantly according to the geographical and clinical setting. In low-income countries, 12–35% of children with acute diarrhoea may experience a prolonged course, with 5–7% of cases lasting more than 14 days.<sup>72</sup> Children with underlying chronic medical conditions are at an increased risk of PD due to factors such as intestinal mucosal susceptibility, dietary practices, and concurrent treatments. Parasitic infections (e.g., *Giardia*) may affect up to 20% of children with chronic illnesses or immunodeficiency.<sup>73</sup>

PD is a major cause of morbidity and mortality in unprivileged areas due to the high risk of dehydration, weight loss and perpetuation of a vicious cycle of malnutrition. Affected children often develop stunting and malnutrition, which in turn trigger persistent diarrhoea—further reinforcing the cycle.

### 11.2 Lactose intolerance

Lactose intolerance may occur in infants and young children with acute diarrhoea, although the clinical consequences are usually limited except in severely affected children. Lactose intolerance is a distinct condition from cow milk protein allergy (CMPA), which is immune-mediated and causes varying degrees of injury to the intestinal mucosa. Lactose intolerance is typically transient, self-limiting, and resolves within 2 weeks.<sup>74</sup> However, in certain at-risk infants (e.g., those younger than 3 months or who are malnourished) with infectious diarrhoea, lactose intolerance may significantly alter the course of illness. In the presence of signs suggestive of lactose intolerance—such as perianal redness or acidic stools—a two-week trial of lactose-free milk may be considered.

### 11.3 Diarrhoea in undernourished children

The association between undernutrition and diarrhoeal mortality is bidirectional. The interplay between diarrhoea, poor growth, and adverse neurocognitive development outcomes in young children is well established.<sup>75,76</sup> Undernutrition following diarrhoeal illness results from anorexia, reduced absorptive function, mucosal damage, and nutrient depletion associated with each episode of diarrhoea.<sup>77</sup> A significant proportion of global undernutrition is attributed to enteric infections.<sup>77</sup> Although undernourished children are at higher risk of developing recurrent and complicated AGE, only a few interventions have been studied, with variable results. For example, while probiotics are considered standard practice in otherwise healthy children, their effectiveness in undernourished children remains inconsistent.<sup>78</sup>

### 11.4 Refeeding syndrome

Refeeding syndrome (RS) is a metabolic complication that arises as a result of the rapid reintroduction of nutrition in children who are severely undernourished, have experienced prolonged starvation, or are metabolically stressed due to severe illness. It is characterized by electrolyte imbalances and a sudden, rapid deterioration of clinical condition. RS is more common in children, with the above risk factors, who are rapidly refed, either orally or enterally.<sup>79</sup> The metabolic disturbances of RS include hypophosphataemia, hypokalaemia, hypomagnesaemia, and thiamine deficiency. It is a potentially life-threatening condition.<sup>80</sup>

To prevent RS, it is important to closely monitor the resumption of feeding, particularly in severely undernourished children. Management should begin with small amounts of low-calorie fluids, gradually increasing both the calorie content and volume of feeds over several days. Careful monitoring of electrolyte levels is essential, and prophylactic supplementation—such as with thiamine—may be required to correct or prevent imbalances.

A detailed description of RS management is beyond the scope of this guideline.

### 11.5 Cow's milk protein allergy

Food protein allergy is a potentially serious complication following AGE. A detailed discussion on the management of cow's milk protein allergy (CMPA) following AGE is beyond the scope of this guideline. Children suspected of having CMPA should be referred to a specialist with expertise in paediatric allergy or gastroenterology.

### 11.6 Diarrhoea in immunocompromised children<sup>55, 81, 82</sup>

Diarrhoea is a challenging complication in immunocompromised children and often contributes to significant morbidity and prolonged hospitalisation. Immunocompromised states can arise from primary immunodeficiencies, malignancies, organ transplantation, secondary immunodeficiencies or immunosuppressive therapies.

#### **11.6.1 Epidemiology and aetiology**

The prevalence of diarrhoea in immunocompromised children varies depending on the underlying condition. Infectious causes are predominant, with viral (e.g., norovirus, rotavirus, cytomegalovirus), bacterial (*Clostridium difficile*, *Salmonella spp.*, *Escherichia coli*), and parasitic (e.g., *Cryptosporidium*, *Giardia*) enteropathogens being the most common. The likely pathogen may differ according to the type of primary or secondary immunodeficiency. In cases of primary immunodeficiency, consultation with a clinical immunologist is recommended. Non-infectious aetiologies such as graft-versus-host disease (GVHD) in transplant patients and drug-induced diarrhoea also contribute significantly.

#### **11.6.2 General principles**

##### **Management Strategies**

Management should be tailored to the underlying aetiology and the patient's immunocompromised state. Key components include:

- **Fluid and electrolyte replacement:** Prompt rehydration using oral or intravenous fluids, depending on severity.
- **Antimicrobial therapy:** Empirical or targeted treatment with antibiotics, antivirals, or antiparasitic agents based on diagnostic findings.
- **Nutritional support:** Early initiation of nutritional therapy to prevent or address undernutrition.
- **Immunomodulatory therapy:** Adjustment of immunosuppressive regimens where clinically appropriate.
- **Infection control measures:** Implementation of isolation precautions and strict hygiene protocols to prevent nosocomial transmission.

Complications such as persistent diarrhoea, malnutrition, and sepsis are common when management is delayed or inadequate.

### 11.7 Antibiotic-associated diarrhoea

Antibiotic-associated diarrhoea (AAD) is defined as diarrhoea that occurs during or following antibiotic therapy, after other potential aetiologies have been excluded. It may develop during antibiotic treatment, shortly after cessation, or even several weeks later.<sup>83</sup> AAD is a common complication, reported in both outpatient and inpatient settings, and may affect up to one-third of patients treated with antibiotics.<sup>84</sup>

#### 11.7.1 Pathophysiology

- The exact mechanism underlying AAD is not fully understood. It is likely multifactorial and involves disruption of the gut microbiota by antibiotics, leading to an overgrowth of pathogenic organisms such as *Clostridioides difficile*.
- Altered gut metabolic activity.
- Direct effects of antibiotics on the intestinal mucosa.

### **11.7.2 Risk factors**

Certain factors increase the likelihood of developing AAD:

- Use of broad-spectrum antibiotics such as amoxicillin-clavulanate, cephalosporins, and clindamycin. In general, AAD is more likely to occur with any antibiotics that are poorly absorbed and active against anaerobes
- Young age (<2 years old)<sup>85</sup>
- Prolonged or multiple courses of antibiotics
- Hospitalisation and underlying chronic conditions such as inflammatory bowel disease, cystic fibrosis, or cancer<sup>86</sup>

### **11.7.3 Clinical features**

The spectrum of symptoms includes watery diarrhoea (up to 10 stools per day), abdominal pain, fever, and occasionally bloody diarrhoea.

Severe *C. difficile* infection:

- Profuse diarrhoea accompanied by systemic signs such as fever, rigors, severe abdominal pain, tenderness, or distention, or requiring hospitalisation
- Elevated white blood cell count (>15,000 cells/ $\mu$ L)
- Elevated age-adjusted serum creatinine levels

### **11.7.4 Diagnosis**

The primary symptom of a *Clostridioides difficile* infection is the sudden onset of watery diarrhoea, typically occurring during or shortly after antibiotic treatment. Differentiating *C. difficile* diarrhoea from more common paediatric causes—such as norovirus or rotavirus—can be difficult based on symptoms alone. However, the following clinical and laboratory indicators may raise suspicion that *C. difficile* is the causative pathogen (see Table 10).



**Table 10: Features suggestive of *Clostridioides difficile* infection in children<sup>87</sup>**

| Clinical features                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Severe diarrhoea (large volume, frequency &gt;6 stools/day, abdominal pain)</li> </ul>                                                                                                       |
| <ul style="list-style-type: none"> <li>• Fever (<math>\geq 38.9^{\circ}\text{C}</math>)</li> </ul>                                                                                                                                    |
| <ul style="list-style-type: none"> <li>• &gt;15,000 white blood cells/<math>\mu\text{L}</math></li> </ul>                                                                                                                             |
| <ul style="list-style-type: none"> <li>• Presence of faecal leucocytes or occult blood</li> </ul>                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Hypoalbuminaemia</li> </ul>                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>• Acute abdomen with absence of stool passage due to ileus</li> </ul>                                                                                                                          |
| <ul style="list-style-type: none"> <li>• Radiological evidence of colitis</li> </ul>                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>• Pseudomembranous colitis (severe inflammation of the intestinal mucosa), confirmed by visualisation of pseudomembranes on endoscopy or histologic examination of biopsy specimens</li> </ul> |

**Table 11: Predisposing conditions in children developing *Clostridioides difficile* infection<sup>88</sup>**

|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| Antibiotics exposure within the past 10 weeks                                                   |
| Recent hospitalisation                                                                          |
| Use of acid-suppressing agents (proton pump inhibitors or H <sub>2</sub> -receptor antagonists) |
| Gastrointestinal tube feedings                                                                  |
| Immunodeficient states                                                                          |
| Cystic fibrosis                                                                                 |
| Hirschsprung enterocolitis                                                                      |
| Structural or postoperative intestinal disorders                                                |

Adapted from *Clostridioides difficile* (formerly *Clostridium difficile*). In: Red Book: 2024–2027 Report of the Committee on Infectious Diseases, 33rd ed, Kimberlin DW, Bannerjee R, Barnett ED, Lynfield R, Sawyer MH (Eds), American Academy of Pediatrics, Itasca, IL 2024. p.313.

Laboratory diagnosis of *Clostridioides difficile* infection in children is challenging.<sup>89</sup> A positive test result does not necessarily confirm clinical disease. Tests that identify the organism, such as nucleic acid amplification tests (NAATs), are highly sensitive but cannot distinguish between colonisation and active infection. Tests that detect toxins—such as enzyme immunoassays—are more specific but lack sufficient sensitivity for reliable diagnosis. Moreover, the presence or absence of toxin does not correlate with disease severity.<sup>90</sup>

**11.7.5 Management**

1. Management depends on the severity of diarrhoea:
  - Mild AAD: Discontinue the causative antibiotic if feasible. Provide supportive care, including hydration and close monitoring.
  - Severe AAD or *C. difficile* infection:
    - » Discontinue the implicated antibiotic.
    - » Initiate targeted antibiotic therapy for *C. difficile* (e.g., metronidazole or vancomycin).
    - » Hospitalisation may be necessary for fluid replacement, electrolyte management, and clinical monitoring.
2. Severe *C. difficile* infection requires prompt intervention. Aggressive diagnostic and therapeutic measures are essential. Surgical consultation should be sought early to assess the need for surgery.
3. Probiotics may reduce the incidence and duration of AAD (by approximately one day) through restoration of gut microbiota.<sup>91</sup> They are generally well-tolerated but should be used with caution in immunocompromised or critically ill patients due to rare reports of severe infections.

**Recommendations:**

There is sufficient evidence to support the use of probiotics during antibiotic therapy for high-risk children to reduce the incidence and duration of antibiotic-associated diarrhoea. The following strains may be considered:

- *Lactobacillus rhamnosus* GG
- *Saccharomyces boulardii* (at doses  $\geq 5$  billion CFU/day)

## 11.8 Economic impact of childhood acute diarrhoeal disease in Malaysia

Acute diarrhoeal diseases remain one of the leading causes of hospitalisation among paediatric patients in Malaysia. The burden is greatest among children from lower-socioeconomic backgrounds, where inadequate sanitation and limited access to clean water contribute to higher incidence rates.<sup>92</sup>

### **11.8.1 Direct healthcare costs and loss of productivity**

The average cost of childhood diarrhoea in low- and middle-income countries is estimated at US\$36.56 per outpatient episode and US\$159.90 per inpatient episode. Direct medical expenses—including hospitalisation, physician consultations, medications, and diagnostic tests—account for approximately 79% of total direct costs.<sup>93</sup>

Although rotavirus vaccines are available in Malaysia, cost remains a significant barrier for many families, as the vaccine is not currently included in the national immunisation programme. Parents often miss work to care for children with gastroenteritis, with reported work absenteeism ranging from 1.5–5.9 days, depending on the severity and context of the infection.<sup>94</sup>

### **11.8.2 Rotavirus impact and vaccine cost-effectiveness in Malaysia**

Several surveillance studies have shown that viral gastroenteritis—particularly due to rotavirus—accounts for nearly 40% of childhood diarrhoea cases requiring medical attention in Malaysia.<sup>16, 95</sup> The introduction of the rotavirus vaccine has contributed to a decline in severe cases; however, outbreaks continue to occur, especially in urban slums and rural communities.

Rotavirus gastroenteritis is estimated to cause approximately 27 deaths, 31,000 hospitalisations, 41,000 outpatient visits, and 145,000 cases of home-treated illness annually in Malaysia. This translates to an annual incidence of 1 death per 100,000 children and 12 hospitalisations, 16 outpatient visits, and 57 home-treated episodes per 1,000 children under five years. The annual

economic burden is substantial, with costs estimated at approximately USD\$34 million to healthcare providers and USD\$50 million to society. Productivity losses account for nearly one-third of total societal costs.<sup>96</sup>

Rotavirus vaccines (Rotarix® and RotaTeq®) at tender prices have been deemed cost-effective in Malaysia when assessed using WHO's cost-effectiveness thresholds.<sup>97</sup>

### **Recommendations:**

- Rotavirus vaccination is a cost-effective, equitable, and necessary intervention for reducing the burden of childhood diarrhoeal disease in Malaysia.
- Incorporating rotavirus vaccination into the national immunisation programme could substantially reduce severe gastroenteritis cases, related hospitalisations, and associated healthcare costs.
- High-risk populations—including infants from lower-income households and those with limited access to healthcare—should be prioritised for vaccine coverage.

## **Conclusion**

Acute infectious diarrhoea remains a significant burden among children in Malaysia and is a preventable cause of childhood mortality. Efforts must be strengthened to improve the management of children presenting with acute diarrhoea. It is hoped that this guideline will assist healthcare providers in Malaysia in delivering appropriate care for children with acute diarrhoea, ultimately contributing to improved health outcomes.

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